Double Data Encryption Standard (2DES)

Aim

To implement the **Double Data Encryption Standard (2DES)** algorithm for encrypting and decrypting plaintext using **two different secret keys**, thereby improving security over the standard DES algorithm.

Algorithm

Encryption Algorithm

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the first secret key (K1).

Step 4: Read the second secret key (K2).

Step 5: Encrypt the plaintext using the first key (K1) with DES.

$$C_1 = DES_{K1}(P)$$

Step 6: Encrypt the intermediate ciphertext again using the second key (K2).

$$C = DES_{K2}(C_1)$$

Step 7: Display the final ciphertext.

Step 8: For decryption, first decrypt the ciphertext using the second key (K2).

$$C_1 = DES_{K2}^{-1}(C)$$

Step 9: Decrypt the intermediate ciphertext using the first key (K1).

$$P = DES_{K1}^{-1}(C_1)$$

Step 10: Display the original plaintext.

Step 11: Stop the program.

C Program (Simplified 2DES Demonstration)

```
#include <stdio.h>

#include <string.h>

int main()

{

    char plaintext[100], key1[100], key2[100];

    char cipher1[100], cipher2[100], decrypted[100];
```

```c
int i, len;
printf("Enter Plaintext: ");
scanf("%s", plaintext);
printf("Enter First Key: ");
scanf("%s", key1);
printf("Enter Second Key: ");
scanf("%s", key2);
len = strlen(plaintext);
// First Encryption using Key1
for(i = 0; i < len; i++)
{
    cipher1[i] = plaintext[i] ^ key1[i % strlen(key1)];
}
// Second Encryption using Key2
for(i = 0; i < len; i++)
{
    cipher2[i] = cipher1[i] ^ key2[i % strlen(key2)];
}
printf("\nEncrypted Cipher Text: ");
for(i = 0; i < len; i++)
{
    printf("%02X", (unsigned char)cipher2[i]);
}
// First Decryption using Key2
for(i = 0; i < len; i++)
{
    cipher1[i] = cipher2[i] ^ key2[i % strlen(key2)];
}
// Second Decryption using Key1
for(i = 0; i < len; i++)
{
    decrypted[i] = cipher1[i] ^ key1[i % strlen(key1)];
```

```
    }
    decrypted[len] = '\0';
    printf("\nDecrypted Plaintext: %s\n", decrypted);
    return 0;
}
```

OUTPUT :